

A.2 Notebook Snapshots

Table A.1 lists the frozen snapshots used in evaluation.

Table A.1: Frozen notebook snapshots used in benchmarks.

Kernel ID	Title	Filename	Used in
113620421	Pakistan housing	kaggle_kernel_113620421.json	Ask 1–2, Code 1–4, Agentic 1
112732919	testing-ol	kaggle_kernel_112732919.json	Ask 3, Agentic 2/4
119598996	housing-final	kaggle_kernel_119598996.json	Agentic 3 (not reported)
113620421	Empty cell fixture	fyp_ask_empty_cell_113620421.json	Ask 4 (excluded from thesis)
113620421	Empty cell fixture	fyp_code_empty_cell_50_113620421.json	Code 4

Snapshots reside under `testing/host/data/notebooks/persistent/`. Each file contains a `cells` array with positional index, type (code or markdown), source text, and optional output fields scraped from the Kaggle editor.

A.3 Metric Definitions

A.3.1 Ask mode metrics

Topic coverage measures the fraction of rubric keywords present in the model reply. For Test 1, keywords include *data loading*, *cleaning*, *feature engineering*, *model training*, and *evaluation*.

Evidence alignment checks whether cited cell indices and variable names appear in the snapshot ground truth.

Hallucination heuristic flags claims about cell content or outputs that cannot be verified from the packed context window.

Contract compliance requires zero `tool_calls` and zero browser writes.

A.3.2 Code mode metrics

Code correctness measures keyword alignment between the generated python block and expected terms (e.g., `XGBRegressor`, `fit`, notebook variable names).